Ankit Panigrahi

Design Engineer

https://ankitpanigrahi78.github.io/portfolio/
Work Experience – 3 Years
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PROFILE

Design Engineer with **3 years'** experience in **Design, Development & Manufacturing of aerospace components** including LRUs, test rigs. Experienced in **Concept Design, Theoretical Stress Analysis**, **CAD drafting & modeling, tubing, ICDs, PDR & CDR reports**, working with Engineering and **Manufacturing teams & Business development teams**.

Naukri : https://www.naukri.com/mnjuser/profile

LinkedIn: https://www.linkedin.com/in/ankit-panigrahi-4b520a209/

EDUCATION

		Pass-Out
B. Tech	GITA Autonomous College, Bhubaneswar	2023
12th	Times Scholars Gurukul, Bhubaneswar	2019
10th	Army Public School, Kanpur	2017

CERTIFICATIONS

- GD&T Fundamentals & MATLAB Training
- Master Certification Course in CAD & CAM
- CATIA Piping, Tubing & Digital Mockups Training – Dassault Systèmes

TECHNICAL SKILLS

Combat Aircraft Fuel System Design and Development	Modeling, Drafting, GD&T, BOM, **System Design**, Process Sheet, Engineering Drawings, **Manufacturing Knowledge,** assembly **and cost reduction**, Product Development, **Report Making (Microsoft Office),** KOP, Piping, Tubing, Aerospace technical documentation **(technical proposals, ICDs), compliance matrix, test procedures (QTP, ATP), BOM management**, Root Cause Analysis (**RCA**), Air Worthiness Certification Plan (**ACP**)
Computer-Aided Design (CAD)	**Catia V5, NX Unigraphics**, SolidWorks, AutoCAD, Creo
Product Life Cycle Management (PLM)	**Enovia**, Teamcenter [Basic]

WORK EXPERIENCE

FCL (Fluid Controls Limited); Aerospace Design Engineer — **Dec 2025 – Present**

Project Experience 2

Quick Disconnecting Coupling, Non-Return Valve, Engine Feed Line Coupling for **AMCA ADA Fuel System** & Quick Disconnecting Coupling for **Boeing 777.**

Responsibilities:

- Prepared **Document** and **Concept Design** for PDR.
- Developed detailed **compliance matrices** covering aerospace quality standards, environmental testing.
- Created **General Assembly (GA)** and developed Models using **SolidWorks**.
- Performed **Theoretical Stress Analysis, Spring, O-Rings and Flow Calculations** to validate design integrity.

CTTC (Ministry of MSME); Junior Project Engineer — **Apr 2023 – Nov 2025**

Project Experience 1

CTTC – TDF Projects

- Completed **3-month intensive training** on aerospace manufacturing workflows, from CAD modeling to testing.

Project Experience 1	**Deputed to ADA (Aeronautical Development Agency) for the Design and Development of LRUs [Line Replaceable Units] & System Design.**
Project Experience1	Retractable Probe, Drogue & Relief Valves (LRUs) for **LCA MK2** & Fuel System Pipeline Routing (Modeling) for **Navy & Airforce Directorates**. **Responsibilities:** • Prepared **LRU-level requirements document** for the Combat Aircraft fuel system, ensuring alignment with **ISO, MIL-STD, Stanag** standards and airworthiness requirements set by **CEMILAC** & **ARDC** (Certifying Agency). • Developed detailed **compliance matrices** covering aerospace quality standards, environmental testing. • Designed and developed **combat aircraft fuel system LRUs** in compliance with **MIL-STD-810H**, supporting **Preliminary Design Reviews (PDRs)**, **Critical Design Reviews (CDRs)**, and indigenous product development initiatives. • Created **Interface Control Documents (ICDs)** and developed using **CATIA** and **Enovia**, ensuring full traceability through the **product lifecycle** also prepared **technical reports, presentations.** • Performed **theoretical stress analysis** using hand calculations to validate structural integrity. • Produced **2D schematic drawings** in AutoCAD and comprehensive **assembly procedures** in line with aerospace documentation standards. • Designed and simulated **3D CAD models** for functional testing, supporting validation activities. • Prepared **Master Drawing Index (MDI)**, **Kit of Parts (KOP)**, and process sheets for aerospace assemblies. • Developed **Digital Mock-Ups (DMU)** & **Bills of Materials (BOMs)** for design validation and integration planning. • Worked closely with senior engineers and reporting officers, supporting all phases of LRU design, manufacturing, integration, and qualification.
	<u>Universal Test Rig for Valves (LRU Development) & Testing</u> **Responsibilities:** • Designed 2D/3D CAD models for aerospace test rigs (fuel) per **ASME SECTION VIII Div 1**. • Prepared and executed **ATPs** for component approval. • Coordinated cross-functional teams for testing (high/low temperature, altitude, and endurance), setup, and integration issue resolution.
PERSONAL SKILLS	
Others	1. Problem Solving. 2. Client-facing Communication. 3. Project Handling. 4. Teamwork & Quick learner. 5. Presentation Skills.